

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based

COURSE NAME: COMPUTER VISION
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COURSE OUTCOME COVERED

CO2: *Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)*

Assessment Tool Weightage: 20%

Scenario-Based Questions

7. Simple Segmentation

An image contains objects clearly distinguishable from the background based on intensity.

- (a) Interpret the scenario and explain segmentation requirements.
- (b) Identify key features enabling separation.
- (c) Apply a suitable segmentation technique.
- (d) Justify your decision logically.
- (e) Present your response clearly.

ANS:

Simple Segmentation (20 Marks)

(a) Interpret the scenario and explain segmentation requirements

Interpretation:

The image contains objects that have **different intensity (brightness)** compared to the background. Since the intensity difference is clear, the objects can be easily separated.

Segmentation Requirement:

- Divide the image into **object** and **background**.
- Assign object pixels to one region and background pixels to another.
- Improve object detection and image analysis.

(b) Identify key features enabling separation

The features that help in separating the object from the background are:

- **Pixel Intensity (Brightness):** Objects are brighter or darker than the background.
- **High Contrast:** Clear intensity difference between object and background.
- **Uniform Background:** Background has similar intensity values.
- **Distinct Object Region:** Object forms a connected area.

(c) Apply a suitable segmentation technique

Threshold-Based Segmentation

A threshold value (T) is selected.

Decision rule:

- If **Pixel Intensity** $\geq T \rightarrow$ Object
- If **Pixel Intensity** $< T \rightarrow$ Background

Example

Original Pixel Values:

```
30 35 40
32 150 160
28 155 165
```

Threshold:

$T = 100$

Segmented Image:

```
0 0 0
0 1 1
0 1 1
```

Where:

- **0 = Background**
- **1 = Object**

(d) Justify your decision logically

Threshold segmentation is chosen because:

- It is simple and easy to implement.
- It provides fast computation.

- It works well when the object and background have a clear intensity difference.
- It requires low computational cost.
- It gives accurate results for high-contrast images.

(e) Present your response clearly

Conclusion

Simple segmentation separates an image into **object** and **background** using pixel intensity. Since the objects are clearly distinguishable, **Threshold-Based Segmentation** is the most suitable technique because it is accurate, efficient, and computationally simple.

Flow Diagram

